MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics
Class: Class X (CBSE)

CBSE CLASS 10 – WEEKLY TEST 01
Chapter 1: Real Numbers

Time: 45 Minutes
MM: 20

Document Type: Official Step-by-Step Marking Scheme **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit theorem/formula declaration, complete prime factorisation or contradiction steps, and boxed final answers with units.
2. In Assertion–Reason questions, truth values of both (A) and (R) as well as whether (R) validly explains (A) are explicitly established.
3. Partial marks are allocated per step in brackets: **[0.5M, 1M, 2M]**. Alternative valid methods receive full credit.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: MCQs & Assertion–Reason

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Q1. The HCF of the smallest prime number and the smallest composite number is: [1]

(a) 1 (b) 2 (c) 4 (d) 8

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (b) 2

Smallest prime number $= 2 = 2^1$, and smallest composite number $= 4 = 2^2$. We know:

Formula Used: $\text{HCF}(a, b) =$ Product of smallest powers of each common prime factor

Common prime factor is 2, with smallest power 1, so $\text{HCF}(2, 4) = \boxed{2}$. [1M]

Q2. The LCM of the smallest two-digit composite number and the smallest composite number is: [1]

(a) 12 (b) 4 (c) 20 (d) 40

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (c) 20

Smallest two-digit composite number $= 10 = 2^1 \times 5^1$, and smallest composite number $= 4 = 2^2$. We know:

Formula Used: $\text{LCM}(a, b) =$ Product of greatest powers of all prime factors involved

Taking highest powers of primes 2 and 5: $\text{LCM}(10, 4) = 2^2 \times 5^1 = 4 \times 5 = \boxed{20}$. [1M]

Q3. If $a = 2^3 \times 3$, $b = 2 \times 3 \times 5$, $c = 3^n \times 5$ and $\text{LCM}(a, b, c) = 2^3 \times 3^2 \times 5$, then the value of n is: [1]

(a) 1 (b) 2 (c) 3 (d) 4

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (b) 2

Given $a = 2^3 \times 3^1$, $b = 2^1 \times 3^1 \times 5^1$, $c = 3^n \times 5^1$. We know:

Formula Used: Exponent of prime p in $\text{LCM}(a, b, c) = \max(\nu_p(a), \nu_p(b), \nu_p(c))$

Comparing the exponent of prime 3 in $\text{LCM}(a, b, c) = 2^3 \times 3^2 \times 5$: $\max(1, 1, n) = 2 \implies \boxed{n = 2}$. [1M]

Q4. If two positive integers a and b are written as $a = x^3y^2$ and $b = xy^3$, where x, y are prime numbers, then $\text{HCF}(a, b)$ is: [1]

(a) xy (b) xy^2 (c) x^3y^3 (d) x^2y^2

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (b) xy^2

Given $a = x^3y^2$ and $b = x^1y^3$ with primes x, y. We know:

Formula Used: $\text{HCF}(a, b)$ = Product of smallest powers of common prime factors

Minimum exponent of x is $\min(3, 1) = 1$, and of y is $\min(2, 3) = 2$. Hence $\text{HCF}(a, b) = x^1y^2 = \boxed{xy^2}$. [1M]

Q5. Assertion (A): The HCF of 11 and 17 is 1.
Reason (R): If p and q are distinct prime numbers, then their highest common factor is always 1. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) Both (A) and (R) are true and (R) is the correct explanation of (A).

We know the co-primality property of distinct primes:

Formula Used: Distinct primes $p \neq q$ share only 1 as common factor $\implies \text{HCF}(p, q) = 1$

Since 11 and 17 are distinct prime numbers, their only common factor is 1, so $\text{HCF}(11, 17) = 1$. Thus both (A) and (R) are true and (R) correctly explains (A). [1M]

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SECTION B: Very Short Answer Type — MATHS MANTHAN | (1 × 2 = 2 Marks)

Q6. Explain why $7 \times 11 \times 13 + 13$ and $7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 + 5$ are composite numbers. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

We know the Fundamental Theorem of Arithmetic characterization of composite numbers:

Formula Used: A composite number $n > 1$ has at least one factor other than 1 and n itself.

(i) First expression: $7 \times 11 \times 13 + 13$: Taking 13 common from both terms:

$$7 \times 11 \times 13 + 13 = 13 \times (7 \times 11 + 1) = 13 \times (77 + 1) = 13 \times 78 = 2 \times 3 \times 13^2.$$

Because it is expressed as a product of primes $(2, 3, 13)$ and has factors besides 1 and itself, $\boxed{7 \times 11 \times 13 + 13 \text{ is composite.}}$ [1M]

(ii) Second expression: $7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 + 5$: Taking 5 common:

$$7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 + 5 = 5 \times (7 \times 6 \times 4 \times 3 \times 2 \times 1 + 1) = 5 \times (1008 + 1) = 5 \times 1009.$$

Since it has factors $1, 5, 1009$, and itself (5045), $\boxed{7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 + 5 \text{ is composite.}}$ [1M]

SECTION C: Short Answer Type — MATHS MANTHAN | (3 × 3 = 9 Marks)

Q7. Prove that $\sqrt{2}$ is an irrational number. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Assumption: Suppose, on the contrary, that $\sqrt{2}$ is rational. Then there exist integers a and b $(b \neq 0)$ with $\text{HCF}(a, b) = 1$ (co-prime) such that:

$$\sqrt{2} = \frac{a}{b} \implies a = b\sqrt{2}.$$

Squaring both sides:

$$a^2 = 2b^2. \qquad \text{— (1)} \qquad [1\text{M}]$$

We know the Fundamental Theorem of Arithmetic divisibility theorem:

Formula Used: If a prime p divides a^2, then p also divides a.

From (1), 2 divides a^2; since 2 is prime, 2 divides a. Hence $a = 2c$ for some integer c. [1M]
Substituting $a = 2c$ back into (1):

$$(2c)^2 = 2b^2 \implies 4c^2 = 2b^2 \implies b^2 = 2c^2.$$

Thus 2 divides b^2, so 2 also divides b. Therefore 2 is a common factor of both a and b, contradicting our assumption that a and b are co-prime $(\text{HCF}(a, b) = 1)$.

This contradiction arose from assuming $\sqrt{2}$ is rational. Hence $\boxed{\sqrt{2} \text{ is an irrational number.}}$ [1M]

Q8. Prove that $\sqrt{3}$ is an irrational number. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Assumption: Suppose, on the contrary, that $\sqrt{3}$ is rational. Then there exist integers a and b ($b \neq 0$) with $\text{HCF}(a, b) = 1$ (co-prime) such that:

$$\sqrt{3} = \frac{a}{b} \implies a = b\sqrt{3}.$$

Squaring both sides:

$$a^2 = 3b^2. \qquad \text{— (1)} \qquad \textbf{[1M]}$$

We know the Fundamental Theorem of Arithmetic divisibility theorem:

Formula Used: If a prime p divides a^2, then p also divides a.

From (1), 3 divides a^2; since 3 is prime, 3 divides a. Hence $a = 3c$ for some integer c. **[1M]**
Substituting $a = 3c$ back into (1):

$$(3c)^2 = 3b^2 \implies 9c^2 = 3b^2 \implies b^2 = 3c^2.$$

Thus 3 divides b^2, so 3 also divides b. Therefore 3 is a common factor of both a and b, contradicting our assumption that a and b are co-prime ($\text{HCF}(a, b) = 1$).

This contradiction arose from assuming $\sqrt{3}$ is rational. Hence $\boxed{\sqrt{3} \text{ is an irrational number.}}$ **[1M]**

Q9. Prove that $\sqrt{5}$ is an irrational number. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Assumption: Suppose, on the contrary, that $\sqrt{5}$ is rational. Then there exist integers a and b ($b \neq 0$) with $\text{HCF}(a, b) = 1$ (co-prime) such that:

$$\sqrt{5} = \frac{a}{b} \implies a = b\sqrt{5}.$$

Squaring both sides:

$$a^2 = 5b^2. \qquad \text{— (1)} \qquad \textbf{[1M]}$$

We know the Fundamental Theorem of Arithmetic divisibility theorem:

Formula Used: If a prime p divides a^2, then p also divides a.

From (1), 5 divides a^2; since 5 is prime, 5 divides a. Hence $a = 5c$ for some integer c. **[1M]**
Substituting $a = 5c$ back into (1):

$$(5c)^2 = 5b^2 \implies 25c^2 = 5b^2 \implies b^2 = 5c^2.$$

Thus 5 divides b^2, so 5 also divides b. Therefore 5 is a common factor of both a and b, contradicting our assumption that a and b are co-prime ($\text{HCF}(a, b) = 1$).

This contradiction arose from assuming $\sqrt{5}$ is rational. Hence $\boxed{\sqrt{5} \text{ is an irrational number.}}$ **[1M]**

Q10. CASE STUDY 01: SCHOOL READING CAMPAIGN LIBRARY BOOK DISTRIBUTION [4 Marks]

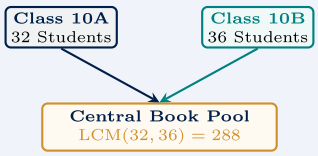
To foster independent reading habits among Class 10 students, a school administration establishes a dedicated classroom library. Class 10 has two sections: Section A with 32 students and Section B with 36 students. The head librarian orders books from a central warehouse and wants to acquire a minimum quantity of books such that they can be distributed completely and equally among the students of Section A or Section B without any books remaining.

(i) Express the student strengths 32 and 36 as products of their prime factors. [1]

(ii) Determine the minimum number of books the librarian must acquire to fulfill the condition. [2]

(iii) If 720 books are acquired instead, how many books does each student of Section A and Section B receive respectively? [1]

▶ CASE-BASED SOLUTION & MARKING SCHEME: [Total: 4 Marks]



Problem Translation: The minimum common stock divisible by both 32 and 36 is LCM$(32, 36) = 288$ books. For Section B, $288 \div 36 = 8$ books/student; for Section A, $288 \div 32 = 9$ books/student.

We know the LCM formula by prime factorisation:

Formula Used: LCM(a, b) = Product of highest powers of all prime factors

(i) Prime factorisations: Splitting 32 and 36 into prime powers:

$$32 = 2 \times 2 \times 2 \times 2 \times 2 = \boxed{2^5}, \qquad 36 = 4 \times 9 = \boxed{2^2 \times 3^2}. \qquad [1M]$$

(ii) Minimum number of books: Taking highest powers of primes 2 and 3:

$$\text{LCM}(32, 36) = 2^5 \times 3^2 = 32 \times 9 = \boxed{288 \text{ books}}. \qquad [2M]$$

(iii) Shares for 720 books: Dividing 720 by 32 and 36 separately:

$$\text{Section B: } \frac{720}{36} = \boxed{20 \text{ books each}} \text{ (exact)}, \qquad \text{Section A: } 720 = 32 \times 22 + 16 \implies \frac{720}{32} = 22.5.$$

Thus Section B students get $\boxed{20 \text{ books each}}$, while for Section A (32 does not divide 720 since $288 \nmid 720$), students receive $\boxed{22 \text{ whole books each with 16 leftover}}$ (or 22.5 arithmetically). [1M]

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